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Predicting Successful Sleep Apnea Treatment With Acetazolamide in **Heart Failure** Patients (HF-ACZ)

ClinicalTrials.gov ID NCT01377987

Sponsor David Andrew Wellman

Information provided by David Andrew Wellman, Brigham and Women's Hospital (Responsible Party)

Last Update Posted 2018-06-13

Completed

No longer looking for participants

Researcher View Tab

Trial Contacts

Contacts	ICMJE
Contact information is displayed when a study is open for participants to join.	

Study Record Dates

First Submitted	ICMJE
2011-06-20	

First Posted ICMJE

2011-06-22

Results First Submitted ICMJE

2018-04-17

Results First Posted ICMJE

2018-06-13

Last Update Posted

2018-06-13

Last Verified *

2018-05

Outcome Measures

Change History

See all versions of this study

Primary (Current) * ICMJE

(Submitted: 2018-05-16)

- The Severity of Sleep Disordered Breathing (Apnea-hypopnea Index, AHI) [Time Frame: 1 week]
  - The frequency of apneas and hypopneas (apnea-hypopnea index) was assessed. The primary measure was the value for non-REM supine sleep. A higher value indicates more severe sleep apnea. A value above 15 indicates the presence of moderate-to-severe sleep apnea.

Primary (Original) * ICMJE

(Submitted: 2011-06-20)

- The severity of sleep disordered breathing (apnea-hypopnea index, nocturnal arterial oxygenation, sleep efficiency and quality, and sleepiness) [Time Frame: 1 week]

Secondary (Current) [*] ICMJE

(Submitted: 2018-05-16)

- Ventilatory Chemoreflex Sensitivity, "Loop Gain" Using Carbon Dioxide Pulses [Time Frame: 1 week]
  - Chemoreflex "loop gain" was assessed according to Sands SA et al AJRCCM 2017 Jan 15;195(2):237-246. Loop gain is a unitless ratio measure that describes the magnitude of the increase in ventilation that occurs in response to a prior reduction in ventilation ("disturbance") and has units of L/min per

L/min. A larger value indicates a more sensitive and unstable control system predisposing to oscillatory breathing. Loop gain was measured on the time scale of 1 min (i.e. response to a 1 cycle/min sinusoidal disturbance, referred to as "LG1"). The procedure involved brief administration of 7% carbon dioxide in air for 0.5 min ("pulses"); tests were repeated every 3 min for 30 min while measuring ventilation and carbon dioxide levels at the nose with patients awake and supine.

measured using 0.5 min pulses of carbon dioxide.

- Sympathetic Activity (Urinary Norepinephrine) [Time Frame: 1 week]
  - Urinary norepinephrine levels overnight
- Left-atrial Volume [Time Frame: 1 week]
  - Left-atrial volume index, echocardiography, bi-plane method. Lower values were considered a favorable outcome. We considered values ≤28 mL/m^2 to indicate normal left atrial volume. Values indicating graded left atrial enlargement were described as follows: mild (29-33 mL/m^2), moderate (34-39 mL/m^2), severe (≥40 mL/m^2).
- Brain Natriuretic Peptide (NT-proBNP) [Time Frame: 1 week]
  - Brain natriuretic peptide (NT-proBNP) in morning
- Pittsburgh Sleep Quality Index [Time Frame: 1 week]
  - Pittsburgh Sleep Quality Index is a measure of self-reported sleep quality containing 19 questions that make up 7 component scores that are added to provide a total score. Total scores range from 0-21 (units on a scale) with higher scores representing reduced sleep quality. A score of 5 or more is interpreted as reduced sleep quality. The total score is reported.

Secondary (Original) [*] ICMJE

(Submitted: 2011-06-20)

- Ventilatory stability, i.e. loop gain, ventilatory responses to carbon dioxide [Time Frame: 1 week]
- Sympathetic activity, neurohormonal activation, and other biomarker indices of heart failure severity [Time Frame: 1 week]
- Cardiac function, assessed by echocardiography [Time Frame: 1 week]
- Respiratory system conductance, reactance, closing volume [Time Frame: 1 week]

Other Pre-specified (Current)

Not provided

Other Pre-specified (Original)

Not provided

Trial Description

Brief Title * ICMJE

Predicting Successful Sleep Apnea Treatment With Acetazolamide in **Heart Failure** Patients

Official Title *§ ICMJE

Physiological Study to Predict Successful Sleep Apnea Treatment With Acetazolamide in **Heart Failure** Patients

Brief Summary *

The ultimate goal is to improve our understanding of the pathophysiology and resistance to effective treatment of sleep disordered breathing in patients with heart failure, with a focus on selecting patients that will benefit specifically from acetazolamide treatment.

The study addresses three primary hypotheses: 1) Acetazolamide treatment will reduce the apnea-hypopnea index and improve markers of heart-failure severity in heart-failure patients with sleep apnea. 2) Acetazolamide will provide the greatest improvement in patients with the most severe ventilatory control instability (strongest chemoreflex response to carbon dioxide; highest loop gain). 3) Acetazolamide will act primarily via stabilizing ventilatory control (reducing loop gain), rather than via improvement to upper airway anatomy, pulmonary congestion, and cardiac function.

Detailed Description

Not provided

Study Type * ICMJE

Interventional

Study Phase * ICMJE

Not Applicable

Study Design *§ ICMJE

Allocation
Randomized

Interventional Model
Crossover Assignment

Masking
Double (Participant, Investigator)

Primary Purpose
Basic Science

Condition * ICMJE

- **Heart Failures**

Intervention * ICMJE

- Drug: Acetazolamide
  - 4 mg/kg, once daily before bed, for 7 days
  - Other Names:
    - Diamox
- Drug: Placebo
  - 4 mg/kg, once daily before bed, for 7 days

Study Arms * ICMJE

- Experimental: Acetazolamide
  - Interventions:
    - Drug: Acetazolamide
- Placebo Comparator: Sugar pill
  - Interventions:
    - Drug: Placebo

Publications

(Includes general and study results' publications, and Pubmed publications referencing this study by ClinicalTrials.gov Identifier (NCT Number))

Not provided

Recruitment Information

Recruitment Status * ICMJE

Completed

Enrollment (Actual) *§ ICMJE

(Submitted: 2017-01-04)

29

Original Enrollment (Estimated) *§ ICMJE

(Submitted: 2011-06-20)

85

Study Start Date *§ ICMJE

2011-08

Primary Completion Date (Actual) *

2016-12 (Final data collection date for primary outcome measure)

Study Completion Date (Actual) *§ ICMJE

2016-12

Eligibility Criteria * ICMJE

Inclusion Criteria (Heart failure patients)

- Left ventricular ejection fraction (EF) <50%, or heart failure with preserved EF
- Age 18-89

Exclusion Criteria:

- severe obstructive respiratory disease
- unstable heart failure status
- recent use of positive airway pressure therapy
- current use of opioids, benzodiazepines
- severe kidney disease
- severe anemia

Sex/Gender * ICMJE

Sexes Eligible for the Study:
All

Ages * ICMJE

18 Years to 89 Years (Adult, Older Adult)

Accepts Healthy Volunteers *§ ICMJE

No

Location Countries ICMJE

United States

Removed Location Countries

Administrative Information

NCT Number ICMJE

NCT01377987

Other Study ID Numbers [*] ICMJE

2011-P-000049/1

Has Data Monitoring Committee

No

U.S. FDA-regulated Product *§

Not provided

IPD Sharing Statement ICMJE

Not provided

Current Responsible Party *

David Andrew Wellman, Brigham and Women's Hospital

Original Responsible Party *

Dr. Atul Malhotra, Brigham and Women's Hospital and Harvard Medical School

Current Study Sponsor * ICMJE

David Andrew Wellman

Original Study Sponsor * ICMJE

Brigham and Women's Hospital

Collaborators ICMJE

American Heart Association

National Institutes of Health (NIH)

Investigators ICMJE

Principal Investigator:
Andrew Wellman, MD, PhD
Brigham and Women's Hospital

Principal Investigator:
Atul Malhotra, MD
Brigham and Women's Hospital

PRS Account
Brigham and Women's Hospital

Symbol Legend

*	Required
*§	Required if Study Start Date is on or after January 18, 2017
[*]	Conditionally required
No Symbol	Unmarked fields are optional. If no information is provided, fields will be labeled as "not provided"
ICMJE	Data element required by the International Committee of Medical Journal Editors (https://www.icmje.org/recommendations/browse/publishing-and-editorial-issues/clinical-trial-registration.html) and the World Health Organization ICTRP (https://www.who.int/clinical-trials-registry-platform)